Resonance in a Damped, Driven Oscillator

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Abstract

We derive the steady response of a linear oscillator with viscous damping to a harmonic force and compare the predicted resonance curve with a simulated air-track experiment. This manuscript is the LaTeX twin of the HyperTeX report report.htex, which adds live sliders.

1 Introduction

Resonance appears whenever a system with a natural frequency is driven periodically: in mechanical structures, electrical circuits and the absorption of light by atoms [1]. The simplest model is a mass on a spring with viscous friction, treated in every mechanics text [3, 2]. Section 2 sets up the model and section 3 derives the resonance curve shown in fig. 1.

2 Model

A mass m attached to a spring of stiffness k and a damper with coefficient c, driven by $F_0 \cos \Omega t$, obeys

$$m\ddot{x} + c\dot{x} + kx = F_0 \cos \Omega t. \tag{1}$$

With $\omega_0 = \sqrt{k/m}$ and $\zeta = c/(2\sqrt{mk})$ this becomes

$$\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2 x = \frac{F_0}{m} \cos \Omega t. \tag{2}$$

For $\zeta < 1$ the free solution decays as $e^{-\zeta\omega_0 t}$ while oscillating at $\omega_\mathrm{d} = \omega_0\sqrt{1-\zeta^2}$.

3 Steady response

Substituting $x = \mathrm{Re}\big(\hat{x}\, e^{i\Omega t}\big)$ into eq. (2) and writing $r = \Omega/\omega_0$ gives

$$\frac{|\hat{x}|}{F_0/k} = \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}. \tag{3}$$

Proposition 1 *For $\zeta < 1/\sqrt{2}$ the amplitude (3) peaks at $r = \sqrt{1-2\zeta^2}$ with height $1/\big(2\zeta\sqrt{1-\zeta^2}\big)$.*

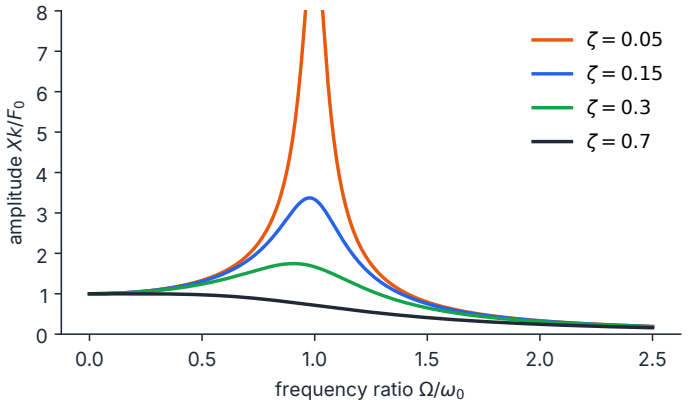


Figure 1: Resonance curves (3) for four damping ratios.

Table 1: Fitted parameters (simulated data).

Quantity	Value	Uncertainty
$\omega_0/2\pi$	1.62 Hz	0.01 Hz
ζ	0.061	0.004
$Q = 1/(2\zeta)$	8.2	0.5

The fitted values in table 1 give a peak height of 8.2, in line with proposition 1. A quality factor this large means the oscillator selects a narrow band of frequencies [2].

References

[1] A. P. French. *Vibrations and Waves*. W. W. Norton, 1971.

[2] L. D. Landau and E. M. Lifshitz. *Mechanics*. Pergamon, 3rd edition, 1976.

[3] John R. Taylor. *Classical Mechanics*. University Science Books, 2005.